MUBAS

SCHOOL OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION SYSTEMS

END OF SEMESTER EXAM

INTERCONNECTING NETWORKS (CIS-INN-211)

(100 Marks)

Lecture: T. Mumba

5th May 2026

Instructions:

Answer all questions.

1. Question one (25 marks)
   a. What is the primary purpose of Wide Area Networks (WANs) in modern day enterprises? (1 mark)
   b. Explain three differences between a Local Area Network (LAN) and a Wide Area Network (WAN). (6 marks)
   c. A business is debating between using a contemporary broadband VPN solution or conventional leased line to connect its branch offices. List at least two benefits and two drawbacks for each of these two WAN technologies. (8 marks)
   d. Explain five factors that a organisation should consider when selecting WAN technology. (10 marks)

2. Question two (25 marks)
   a. Router A and Router B are connected via Serial 0/1/0 interfaces. You are tasked with configuring the link. Write the Cisco IOS commands (including modes) required on Router A to achieve the following:
      i. Enter global configuration mode. (1 mark)
      ii. Assign a hostname to the router. (1 mark)
      iii. Enter interface configuration mode for serial 0/1/0. (1 mark)
      iv. Assign a valid IP address to the interface serial 0/1/0. (2 marks)
      v. Set the encapsulation protocol to PPP. (1 mark)
      vi. Assume Router B is the DCE, set the clock rate on the DCE side to 250000. (1 mark)
   b. Jenkins, a network administrator, recently set up a Point-to-Point Protocol (PPP) connection between two remote locations using Password Authentication Protocol (PAP). During a connection attempt, the router on

site A send the appropriate credentials to the router on site B for authentication. The connection is successful.

i. Explain in detail the security risk associated with PAP authentication? (2 marks)
ii. State two reasons why might PAP still be used despite the associated risk? (2 marks)

c. IPSec is often used to set up VPNs and helps keep data sent over public networks secure.
   i. Explain how IPSec provides security. (6 marks)
   ii. Mention two main protocols used by IPSec to provide security. (2 marks)
   iii. Describe three benefits of using IPSec VPN over dedicated Point-to-Point connections. (6 marks)

3. Question 3 (25 marks)

a. Wilmont University network has 200 switches that require VLAN updates every semester. Currently, engineers configure devices manually.
   i. State any three challenges associated with manual configurations. (3 marks)
   ii. Briefly explain how automation would improve the process above. (2 marks)
   iii. Mention two tools that can be used for automation in the given context. (2 marks)
   iv. Identify one potential risk to be considered and how you can mitigate the risk. (3 marks)

b. Software Defined Networking (SDNs) is becoming popular in modern enterprises. The Open Networking Foundation identifies three main components of Software-Define Networking.
   i. Describe the main components of Software Defined Networking? (9 marks)
   ii. With relevant examples, explain the differences between Southbound and Northbound API. (6 marks)

4. Question 4 (25 marks)

a. Define Network Address Translation (NAT). (3 marks)
b. Compare and contrast static NAT and dynamic NAT, providing a specific use case for each. (4 marks)
c. Zathu limited company has 30 computers using private IP addresses. These computers are used by employees who need to access the internet simultaneously. The company consults an ISP who provides a single public IP address for their use.
   i. What type of network Address Translation (NAT) should the organisation implement and why? (2 marks)

ii. What is the main advantage of implementing the type of NAT you have chosen? (2 marks)

d. A company uses private IP addresses 10.1.1.0/24 for its internal workstations and needs to allow them to access the internet through router R1, which has a public IP address of 209.165.200.2 on its Serial 0/1/0 interface.
   i. Write Cisco IOS commands that you would use to configure a static NAT mapping that maps the internal web server with IP address 10.1.1.50 to the public IP address 209.165.200.2. (7 marks)
   ii. A user reports they cannot access the internet. You suspect a NAT issue. State two commands you would use to verify the NAT configurations. (2 marks)
   iii. Explain the difference between an inbound and outbound ACL. (3 marks)
   iv. Why is the placement of ACLs critical in a network? (3 marks)